ME403 – Introduction to Robotics

Lab 00: Introduction Week Handout

Sabanci University | Spring 2025-26

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Lab Outcomes

By the end of this lab, you will be able to:

- Set up a Python virtual environment for robotic control.
- Connect to the Dobot Magician over USB-serial using the GUI.
- Move individual joints and understand each joint's role and limits.
- Observe how joint angles map to the end-effector position using live pose readout.

1. Overview

This lab introduces the **Dobot Magician** robotic arm. You will set up your development environment, connect to the robot via a **graphical user interface (GUI)**, and explore how joint angles control the arm.

Week 1 Note

MG400 (Ethernet/TCP-IP) exercises are covered in the Week 1 handout (`lab01`). See `Students/01_SecondWeek/` for the MG400 student scripts.

Provided Files

File	Purpose
`00_magician_gui.py`	GUI application: port discovery, connection, alarm clearing, joint control, live pose readout.
`utils.py`	Shared helper functions (safe bounds, port finder, move wrappers).
`requirements.txt`	Python dependencies (install with `pip`).

2. Quick Start Guide

Environment Setup — Python venv

Linux:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r magician/requirements.txt
```

```
cd magician
python 00_magician_gui.py
```

Windows (PowerShell):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r magician\requirements.txt
cd magician
python 00_magician_gui.py
```

3. Hardware Setup

1. **Surface:** Place on a flat, stable surface. Clear a 40 cm radius around the base.
2. **Posture:** Manually set the arm to a neutral posture (both segments at ≈45°) before powering on.
3. **Power:** Connect the 12 V DC adapter. **USB power alone is insufficient.**
4. **Communication:** Connect the USB cable (CP210x USB-to-UART bridge).

Status Signals:

- **Green LED + Single Beep:** Ready.
- **Red LED + Rapid Beeping:** Alarm state — click **Clear Alarms** in the GUI.
- **Yellow/Flashing LED:** Busy (homing or executing a command).

4. OS-Specific Notes

4.1. Linux

Users must have permission to access serial devices. Run the following once:

```
sudo usermod -a -G dialout $USER
# Log out and log back in for changes to take effect.
```

The Magician typically appears as `/dev/ttyUSB0` or `/dev/ttyACM0`.

4.2. Windows

The driver is usually handled by Windows Update. To find the port manually:

1. Open **Device Manager**.
2. Expand **Ports (COM & LPT)**.
3. Look for **Silicon Labs CP210x USB to UART Bridge** (e.g., `COM3`).

The GUI auto-discovers the port, so manual lookup is typically unnecessary.

5. The Dobot Magician

The **Dobot Magician** is a 4-axis desktop robotic arm with:

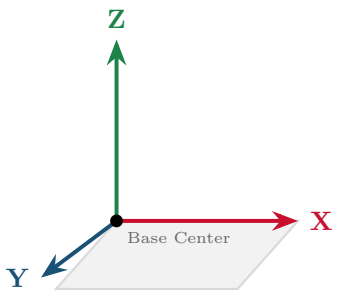
- **Reach:** ≈320 mm maximum.
- **Payload:** 500 g at full extension.
- **Repeatability:** ±0.2 mm.
- **Communication:** USB-serial (CP210x chip).

The arm has two main segments connected by revolute joints:

- **Upper arm (L1):** 135 mm — from the shoulder (J2) to the elbow.
- **Forearm (L2):** 147 mm — from the elbow to the end-effector.

A **parallel linkage** on J3 keeps the forearm's firmware angle relative to the upper arm (body-frame), rather than world-frame absolute.

5.1. Coordinate System



- **X-axis:** Forward (pointing away from base).
- **Y-axis:** Left/Right (Left is positive).
- **Z-axis:** Vertical (Up is positive).
- **Units:** Millimeters (mm).
- **Rotation (R):** Tool rotation in degrees (°).

5.2. Joint Definitions and Limits

Each joint controls a different degree of freedom. The GUI enforces these limits automatically (values are clamped before sending to the robot).

Joint	Motion	Safe Range	Units
J1	Base Rotation	−90 to 90°	deg
J2	Rear Arm Elevation	0 to 85°	deg
J3	Forearm Elevation	−10 to 85°	deg
J4	End-Effector Rotation	−90 to 90°	deg

Cartesian Dimension	Safe Operating Range	Unit
X (Reach)	120 to 315 mm	mm
Y (Slew)	±158 mm	mm
Z (Height)	5 to 155 mm	mm

6. Using the GUI

1. **Launch:** `python 00_magician_gui.py`
2. **Connect:** Select the port (auto-discovered with a star marker) and click **Connect**. The GUI clears any alarms and homes the robot automatically.

3. **Move Joints:** Use the spinboxes and +/− buttons to set a target angle, then click **Move** for that joint, or **Send All** to move all four joints at once.
4. **Live Pose:** The right panel shows the current X, Y, Z, R and J1–J4 values, updated every 500 ms.
5. **Joint Limits:** The limits panel shows the safe range for each joint.
6. **Details:** The log panel at the bottom shows connection events, alarms, and move commands.

7. Troubleshooting

Common Connection Issues

Symptom	Recommended Action
`Access Denied`	Ensure **DobotStudio** and other Python scripts are closed.
`Device not found`	Verify both USB and Power cables. Click **Refresh** in the GUI to refresh ports.
Motors don't move	Check the 12 V supply. If LED is red, click **Clear Alarms** in the GUI.
Erratic movement	Reset to neutral posture and power cycle the robot.

8. Safety Mandates

Mandatory Safety Rules

1. **Workspace:** Maintain a clean area. No cables in the reach zone (40 cm radius around the base).
2. **Initialization:** Always let the GUI complete its connection sequence (alarm clear + homing) before sending commands.
3. **Joint Limits:** The GUI clamps all values to the safe ranges shown in the Joint Limits panel.
4. **Supervision:** Stay within reach of the power cable at all times.